

Materials and Methods

LD decay and LD-based complexity reduction

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Module 0 (_PK pipeline)

Working draft for the _PK pipeline. It covers marker processing, LD decay and per-marker local LD support, and the two-stage LD-based complexity reduction that yields the shared LD-reduced units. Ancestry sorting and genomic architecture, which use these units, are documented in the Module A methods. Blue boxes mark items that remain open.

S1 Scope and analytical principles

These methods describe LD decay estimation and LD-based complexity reduction, which transform the marker data into the shared set of LD-reduced analytical units used by the downstream ancestry-sorting and genomic-architecture analyses (Module A).

Three principles guided the analyses. First, markers in LD carry partly redundant information. We therefore used clusters of strongly correlated markers as LD-reduced analytical units wherever possible, while retaining marker-level analyses as diagnostics of spatial pseudoreplication. We do not assume that the resulting units are strictly statistically independent. Second, a variable evaluated as a predictor was not also used to select the analysed data. In particular, loci were gated by parental allele frequency rather than by the diagnostic index (DI), leaving the full range of DI available for analysis. Third, these analyses do not assume complete demographic independence of the twenty hybrid populations; analyses that use variation among populations are interpreted as descriptive associations.

Open item. The general sample description currently reports 165 sequenced hybrid individuals, whereas the analysis objects contain 164. Confirm which individual was excluded, most likely because of excessive missing data, and report the identifier and exclusion criterion in the final text. Until then, 164 is used for all analysis-specific counts.

S1.1 Marker processing and ancestry orientation

Genotypes and marker-level ancestry information were obtained from the DIEM output for this data set. Sites were restricted to biallelic markers. DIEM provides a polarity and a diagnostic index for each marker. The diagnostic index measures the degree to which allele frequencies differentiate the parental reference samples. For markers with the opposite polarity, genotype dosages were replaced by $2 - \text{dosage}$, placing dosages on a consistent ancestry-oriented scale across the genome.

Minor allele frequency (MAF) was calculated among hybrid individuals only, and markers with hybrid $\text{MAF} \geq 0.05$ were retained. The resulting analysis set contained 1,114,340 markers. Two matrices were constructed over this same marker set. The hybrids-only matrix was used for LD estimation, clustering, and complexity reduction. A second matrix containing hybrids and parental references was used when parental allele frequencies were required, including ancestry

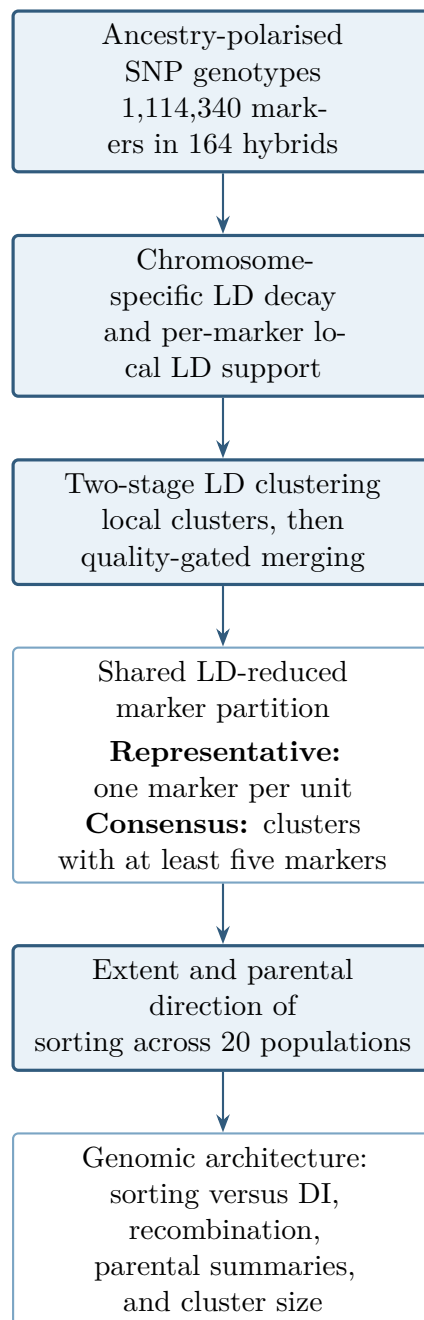


Figure 1: **Analytical workflow (Module A).** LD decay and clustering transform the marker data into a shared set of LD-reduced units represented either by a single marker or, for larger clusters, by a consensus genotype. The same units feed the analyses of the extent, direction and genomic architecture of ancestry sorting.

orientation and marker-level parental summaries. Keeping the parents out of the LD calculations prevents parent–hybrid differentiation from dominating estimates intended to describe LD within the hybrid populations.

S2 LD decay and local LD support

S2.1 Chromosome-specific LD decay

LD decay was modelled separately for each chromosome as

$$r^2(d) = b + \frac{c - b}{1 + ad}, \quad (1)$$

where d is physical distance in base pairs, a is the decay rate, b is long-range background r^2 , and c is short-range r^2 . Background LD was estimated from inter-chromosomal marker pairs. Within each chromosome, the curve was fitted across twenty sliding windows with 50% overlap. Up to 5,000 marker pairs were sampled per window, stratified across twenty distance classes to represent short and long distances comparably.

Within each distance class, LD was summarized by the 0.95 quantile of r^2 to characterize the upper LD landscape, which was the primary interest of this analysis. Parameter estimates were subsequently aggregated over the central 95% of windows to limit the influence of extreme outlier window-level fits (LD-decay supplementary figure).

Only pairs for which both markers had $\text{MAF} > 0.1$ contributed to the background and decay-curve fits. This restriction reduces the frequency-driven depression of r^2 at rare alleles.

Since crossover number generally scales less than proportionally with physical chromosome length, this predicts greater recombination per unit physical distance, and consequently faster LD decay, on shorter chromosomes. A robust regression of a on chromosome size was therefore fitted genome-wide. For a relative decay threshold ρ , defined as the proportion of the fitted short-distance LD that had decayed, the corresponding physical distance was calculated as

$$d = \frac{\rho}{a(1 - \rho)}. \quad (2)$$

This allowed window sizes to reflect each chromosome’s fitted decay behaviour rather than imposing a single physical-distance threshold across the genome.

S2.2 Per-marker local LD support

For each marker, local LD support (ld_w) was calculated as the median r^2 between that marker and neighbouring markers within the window corresponding to $\rho = 0.95$ that denotes the distance at which LD has declined by 95% relative to its fitted short-distance value. Unlike map-based recombination rate, ld_w is measured directly from the genotype data at marker resolution and requires no interpolation.

We evaluated ld_w and the window-level decay rate a as indicators of low recombination using map-based recombination rate as a reference. Across definitions based on the lowest 5%, 10%, and 25% of recombination windows, the area under the receiver-operating-characteristic curve was 0.91–0.98 for a and 0.73–0.86 for ld_w . Accordingly, a was the stronger window-level indicator, whereas ld_w supplied the marker-resolution measure needed to identify clusters for reconsideration during complexity reduction (the recombination-discrimination and local-LD-track supplementary figures).

S3 Two-stage LD-based complexity reduction

S3.1 Terminology and outputs

The procedure has two stages: (i) local clustering with a complete-linkage refinement and (ii) targeted, quality-gated merging. Both stages define a single partition of markers, from which two representations are produced:

- one representative marker per cluster, forming an LD-pruned marker set;
- one expected multi-locus genotype (eMLG) consensus dosage for clusters containing at least five markers.

We use “LD cluster” or “LD-reduced unit” for a set of strongly correlated markers. These units are not haplotype blocks: markers are grouped by the partition of sampled chromosomes that they report, and clusters may therefore overlap in physical position. The procedure reduces redundancy and constrains residual LD but does not make them strictly independent; the remaining dependence is assessed or controlled where it matters downstream.

S3.2 Stage 1: local clustering

For each chromosome, the clustering distance was derived from its fitted LD decay curve at the half-way decay distance $\rho = 0.5$. The threshold was not allowed to vary further within chromosomes, because locally elevated LD is precisely where redundant markers should be grouped more strongly.

Clustering was performed on a sparse LD graph constructed from pairwise LD estimates between markers separated by no more than 400 SNPs, a neighborhood spanning approximately the $\rho = 0.99$ decay distance. Markers were first assigned to connected components of the thresholded graph (single linkage clustering). Each component was then refined by complete-linkage hierarchical clustering using the dense pairwise r^2 matrix within the component. The complete-linkage step limits single-linkage chaining, in which two weakly correlated markers could otherwise be joined through an intermediate marker correlated with both. The representative marker of a cluster was the member with the highest median r^2 to the other cluster members.

S3.3 Stage 2: targeted, quality-gated merging

The finite window used to construct the Stage 1 edge list makes the first stage conservative in extended low-recombination regions: marker pairs outside the window are not evaluated and therefore cannot satisfy complete linkage. Stage 2 reconsidered candidate clusters directly from their genotypes, without the Stage 1 window restriction.

A Stage 1 cluster was carried forward as a candidate for Stage 2 merging if any member had $ld_w > 0.025$ or if the cluster contained at least five markers. This screening reduced unnecessary graph comparisons: small clusters composed entirely of low- ld_w markers were unlikely to show sufficient LD with neighbouring clusters to merge and were therefore left unchanged. The size criterion nevertheless retained larger Stage 1 clusters for evaluation even when none of their individual markers exceeded the local-LD threshold.

Flagged clusters were considered for merging using a distance-restricted, average-linkage dynamic tree cut. A candidate merge was accepted only while both of the following criteria were satisfied:

1. consensus fidelity

$$\text{score}_{\text{eMLG}}(x) = \text{cor}\{\text{round}(x), x\}^2 \geq 0.80;$$

2. a pairwise correlation floor of $r^2 \geq 0.2$ between the sides being merged.

The fidelity criterion evaluates whether the resulting consensus remains close to a discrete genotype. The correlation floor prevents a small, weakly related set of markers from being absorbed into a large cluster whose existing consensus would otherwise dominate the fidelity score. These gates constrain merging but do not establish complete statistical independence among the final units. Observed residual correlations among neighbouring merged units were of order $r^2 \approx 0.3$ after correction for relatedness and were local in genetic distance.

Candidate merges were restricted to clusters separated by no more than 1.0 cM on the genetic map so that the resulting units primarily represented local linkage. Persistent LD across greater recombinational distances is less readily attributable to limited recombination alone and may instead reflect processes such as population structure, admixture history or selection. Such long-range associations were therefore retained as relationships between distinct units rather than used to collapse them into a single cluster.

S3.4 Output and validation

The final partition contained 470,035 LD clusters. Of these, 32,871 contained at least five markers and had an eMLG consensus. Smaller clusters remain valid LD-reduced units and retain their representative marker; the five-marker threshold is therefore a threshold for storing a consensus, not for defining a cluster.

The procedure was evaluated by comparing Stage 1 and combined Stage 1–2 clusterings in extended-LD regions (Stage-1-versus-combined supplementary figure), by examining the distribution of consensus fidelity across ld_w thresholds (consensus-fidelity supplementary figure), and by quantifying the ability of ld_w to identify low-recombination regions. These checks assess whether Stage 2 consolidates fragmentation where long-range LD is present without broadly collapsing weakly correlated markers.

S4 Software, reproducibility, and current status

LD decay estimation, clustering, and consensus construction were implemented using functions from **LDscnR**. Project-level analyses were performed in R using scripts in the **formica_hybrid** repository, folder `module0_ld_pruning/R/`: `module0_LD_decay_from_DIEM.R` (LD decay and per-marker local LD support), `module0_ld_pruning_DIEM.R` (two-stage clustering and consensus construction), and `module0_fig_ld_tracks.R` (local-LD track figures). Shared genotype inputs are read from `data/`; module-specific intermediates and the final clustering are written to `module0_ld_pruning/data/`; figures to `module0_ld_pruning/Figures/`.

Open item. Add the final **LDscnR** citation, version, and repository or archive identifier before submission.

Table 1: Principal parameter settings (Module 0).

Analysis	Parameter	Value
Marker set	Hybrid MAF threshold	≥ 0.05

Analysis	Parameter	Value
	Retained markers	1,114,340
LD decay	Windows per chromosome	20, 50% overlap
	Marker pairs per window	$\leq 5,000$, 20 distance strata
	r^2 summary	0.95 quantile
	Pair MAF for curve fitting	> 0.1 for both members
Stage 1	Relative LD threshold	$\rho = 0.5$, per chromosome
	Clustering	connected components, then complete linkage
	Representative	highest median r^2 to cluster members
Stage 2	Flagging	$ld_w > 0.025$ or ≥ 5 markers
	Consensus fidelity	≥ 0.80
	Pairwise correlation floor	$r^2 \geq 0.2$
	Distance restriction	1.0 cM
	Stored consensus	clusters with ≥ 5 markers
	Final partition	470,035 clusters; 32,871 with consensus